

5(a)	<i>two from</i> protein coat / capsid / capsomeres ; nucleic acid / <u>DNA or RNA</u> (core) ; I single- or double-stranded A 'DNA/RNA' I genetic material size given in nanometres / smaller than prokaryotes ; I small / very small / microscopic I acellular	2
5(b)	<i>three from</i> 1 drugs can be <u>inhibitors</u> (of neuraminidase) ; <i>either</i> 2 (competitive inhibitor) so binds to active site <i>or</i> (non-competitive inhibitor) so binds to allosteric site / AW <i>or</i> (drugs may act by) breaking down / hydrolysing / denaturing, all / part of neuraminidase ; A changes active site 3 no / less, enzyme-substrate complexes formed ; A ESC A substrate not able to bind to enzyme A neuraminidase cannot bind to (host cell) receptor 4 receptor is (still) complementary to haemagglutinin ; 5 haemagglutinin becomes attached to, cell receptor / host cell (so newly formed virus does not leave the cells) ;	3

Question	Answer	Marks
5(c)	<i>four from</i> 1 antigen presentation ; <i>in correct context</i> 2 clonal selection / activation, of specific, B-lymphocytes / T-lymphocytes ; A B cells / T cells 3 (lymphocytes) divide by mitosis / undergo clonal expansion ; 4 B-lymphocytes, differentiate into / mature into / form / AW, plasma cells ; 5 antibodies secreted by plasma cells ; 6 T-helper cells secrete cytokines ; 7 cytokines stimulate / AW, B-lymphocytes / plasma cells / humoral response ;	4

Question	Answer	Marks
5(d)(i)	<i>max 2 (out of total three marks)</i> <i>advantages</i> <i>allow references to other pathogen types</i> - antibodies are provided to people immediately / no delay for plasma cells to secrete antibodies ; A immediate, immunity / protection R immediate (immune) response - antibodies, immediately neutralise toxins / prevent viruses entering cells ; - prevents disease (in the individual) / promotes quicker recovery ; - prevents spread of the pathogen through the population / prevents people dying ; - antibodies can be manufactured quickly in response to mutations that occur in virus / AW ; <i>max 2 (out of total three marks)</i> <i>disadvantages</i> - short-term / temporary (immunity) ; - no memory cells produced ; - can have infections of <u>same</u> pathogen again ; - allergic reaction / immune response, to the (non-human) antibodies given ; <i>ref. to</i> cost qualified ; e.g. needs to be repeated / high cost of production of antibodies - AVP – for advantage (A) or disadvantage (D) ; e.g. (A) passive can be used for people who are malnourished / immunosuppressed e.g. (A) <i>ref. to</i> using a vaccine with a (live) pathogen that might give person the disease	3
5(d)(ii)	across the placenta ; A via umbilical cord in breast milk / colostrum / breast feeding / during lactation ;	2